

Myntra Fashion E-commerce Management System

1) Introduction

The entities required for the Myntra Fashion E-commerce Management System are identified based on the business requirements of an online fashion shopping platform. Each entity is designed with appropriate attributes, primary keys, foreign keys, unique constraints, and validation rules to ensure accurate and efficient data management. The relationships among these entities represent the interactions between customers, products, orders, payments, and other business processes.

This entity analysis forms the foundation for developing the Entity Relationship (ER) diagram, relational schema, normalization process, and database implementation. A well-designed database helps maintain data integrity, minimizes redundancy, improves query performance, and supports efficient management of online shopping operations.

2) Entity Analysis

Entity: Customer

Attribute	Key	Type
Customer_ID	Primary Key	Integer
First_Name	Not Null	Varchar
Last_Name	Not Null	Varchar
Email	Unique Key	Varchar
Phone_Number	Unique	Varchar
Password	Not Null	Varchar
Gender	-	Varchar
Address	Not Null	Text

Entity: Brand

Attribute	Key	Type
Brand_ID	Primary Key	Integer
Brand_Name	Unique Key	Varchar
Country	-	Varchar

Description	-	Text
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Entity: Category

Attribute	Key	Type
Category_ID	Primary Key	Integer
Category_Name	Unique Key	Varchar
Description	-	Text

Entity: Product

Attribute	Key	Type
Product_ID	Primary Key	Integer
Product_Name	Not Null	Varchar
Brand_ID	Foreign Key	Integer
Category_ID	Foreign Key	Integer
Size	-	Varchar
Color	-	Varchar
Price	Not Null	Decimal
Stock	Not Null	Integer
Description	-	Text

Entity: Cart

Attribute	Key	Type
Cart_ID	Primary Key	Integer
Customer_ID	Foreign Key	Integer
Total_Items	-	Integer
Grand_Total	-	Decimal

Entity: Order

Attribute	Key	Type
Order_ID	Primary Key	Integer
Customer_ID	Foreign Key	Integer
Order_Date	-	Date
Order_Status	-	Varchar
Total_Amount	-	Decimal

Entity: Order_Item

Attribute	Key	Type
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OrderItem_ID	Primary Key	Integer
Order_ID	Foreign Key	Integer
Product_ID	Foreign Key	Integer
Quantity	-	Integer
Unit_Price	-	Decimal

Entity: Payment

Attribute	Key	Type
Payment_ID	Primary Key	Integer
Order_ID	Foreign Key	Integer
Payment_Method	-	Varchar
Payment_Status	-	Varchar
Payment_Date	-	Date
Amount	-	Decimal

Entity: Wishlist

Attribute	Key	Type
Wishlist_ID	Primary Key	Integer
Customer_ID	Foreign Key	Integer
Product_ID	Foreign Key	Integer
Date_Added	-	Date

Entity: Review

Attribute	Key	Type
Review_ID	Primary Key	Integer
Customer_ID	Foreign Key	Integer
Product_ID	Foreign Key	Integer
Rating	Check (1-5)	Integer
Review_Text	-	Text
Review_Date	-	Date

Entity: Shipment

Attribute	Key	Type
Shipment_ID	Primary Key	Integer
Order_ID	Foreign Key	Integer
Delivery_Address	Not Null	Text
Shipment_Date	-	Date
Delivery_Date	-	Date

Shipment_Status	-	Varchar
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3) Entity Relationships

Entity 1	Relationship	Entity 2	Cardinality	Participation
Customer	Owns	Cart	One-to-One	Partial / Total
Customer	Places	Order	One-to-Many	Partial / Total
Customer	Adds	Wishlist	One-to-Many	Partial / Total
Customer	Writes	Review	One-to-Many	Partial / Total
Brand	Supplies	Product	One-to-Many	Partial / Total
Category	Contains	Product	One-to-Many	Partial / Total
Cart	Contains	Product	Many-to-Many*	Total / Partial
Order	Includes	Order_Item	One-to-Many	Total / Total
Order_Item	Refers To	Product	Many-to-One	Total / Partial
Order	Has	Payment	One-to-One	Total / Total
Order	Has	Shipment	One-to-One	Total / Total
Product	Receives	Review	One-to-Many	Partial / Total
Wishlist	Stores	Product	Many-to-One	Total / Partial